Multi-agentic workflow for live financial news

Project Summary

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Aim

1. Solving the "Second Order" Analysis Gap

Most investors only use simple AI tools to react to direct news (e.g., "Nvidia stock rose because they launched a chip"). My aim is to bridge the "Reasoning Gap".

Goal: To develop a **Recursive Multi-Agent System** that identifies second-order market beneficiaries through automated supply-chain reasoning.

(e.g., AI boom → More users use chatbots services → energy consumption increases → beneficial for energy companies that provide 24/7 energy sources)

2. Mitigating Information Overload with Watchdogs

In a 24/7 market, significant market moves (sudden rise/drop) often occur without immediate news, leading to "panic selling" or "FOMO buying".

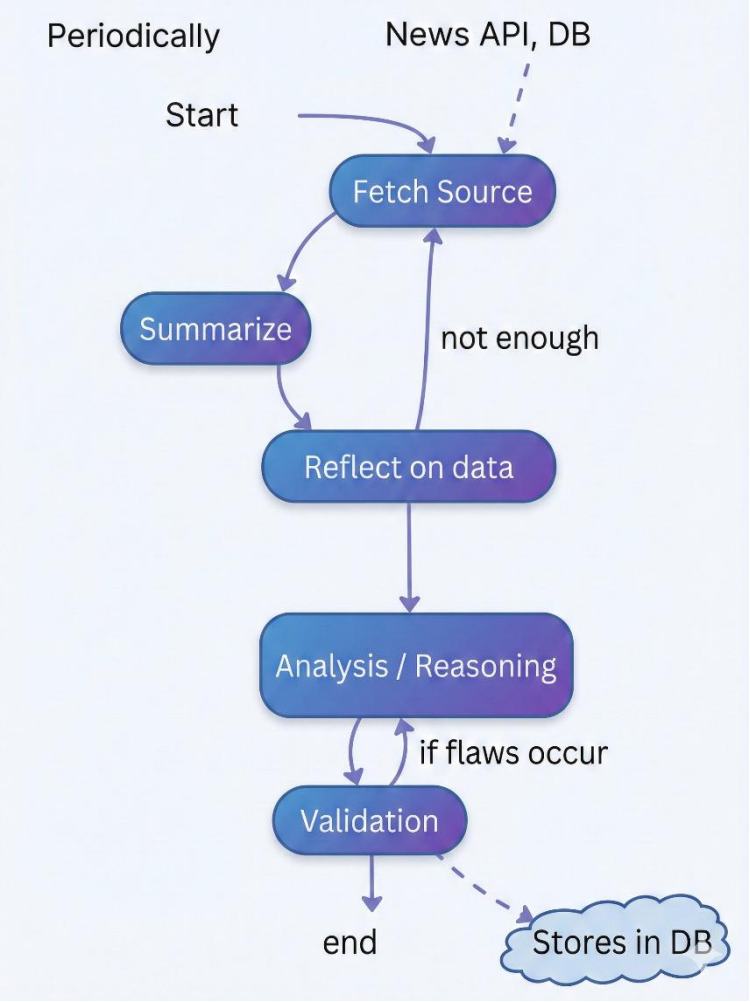
Goal: To implement an **Event-Driven Anomaly Detection** system that automatically triggers deep-dive RAG analysis during periods of high volatility, providing users with instant "Explainable Alpha".

3. Enhancing AI Reliability in High-Stakes Finance

AI hallucinations are a critical barrier to using LLMs for stock trading.

Goal: To create a **Self-Correcting Agentic Workflow** using LangGraph and RAG. This system will validate its own logic against factual historical data stored in a Vector Database to ensure 100% factual grounding for all investment advice.

High-level LangGraph workflow



Start Periodically (e.g. Weekly): Sets the context of users interested areas.

Fetch Source (RAG/News API): Use an agent to decide *which* tools to call (News API vs. SEC Filings).

Summarize: Condenses raw data into actionable insights for the next agent.

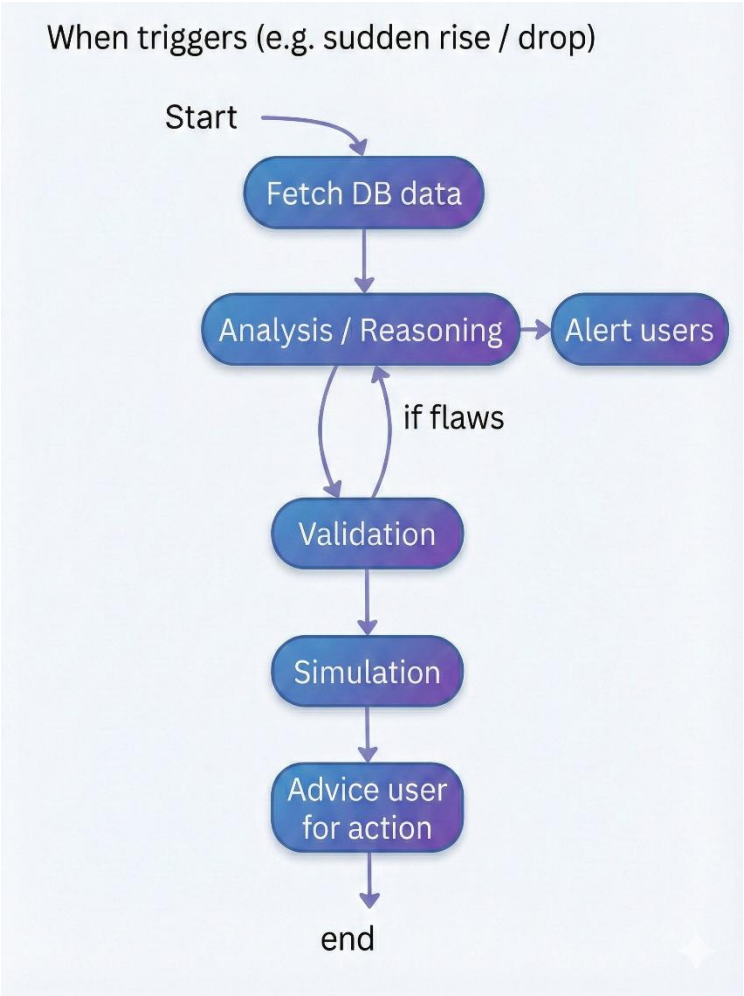
Reflect on Data: Checks for "Information Sufficiency." If data is missing, it triggers a "Re-query".

Analysis: Decomposes the task into sub-steps (e.g., Second-Order logic).

Validation: A separate agent (the "Critic") looks for logical flaws or numerical errors.

Stores all the analysis into DB for further analysis.

Generates a news report on of users interested areas periodically.



Trigger: Uses statistical thresholds (e.g., Z-score > 2 or high-volume rules) to identify "Significant Deviations".

Fetch Data: Pulls the "Now" state from APIs (like Finnhub/Yahoo Finance) and the "Historical" state from the Vector DB to find the "Why" behind the move.

Analysis: Breaks the move into technical vs. fundamental causes (e.g., "Earnings Leak" vs. "Fat Finger Trade").

Validation: Cross-references the reasoning with raw SEC filings or peer-company performance to catch hallucinations.

Simulation: Uses LLM-based agents to simulate "Buy/Sell/Hold" scenarios over 12 months to see the likely trend.

Action: Provides the user with a "High-Conviction Alert," a risk score, and specific "Advice" (e.g., "Hedge with Puts" or "Wait for RSI cooling").

Anti-Hallucination Strategy

Layer 1: Source Grounding (Prevention)

I. Every piece of data fed to the analyst agent must have a source_url and retrieval_timestamp
II. The analyst prompt must include: "Only make claims supported by the provided data. Cite [SOURCE_ID] for every factual claim."

Layer 2: RAG Validation (Detection)

I. After the analyst produces output, the Validator agent queries the vector DB for each claim
II. Uses cosine similarity threshold (e.g., > 0.75) to confirm factual grounding
III. If any claim has no supporting document → flag as UNVERIFIED

Layer 3: Numerical Cross-Check (Verification)

I. Extract all numbers from the analysis (revenue, growth %, margins)
II. Compare against structured data (SEC XBRL, Yahoo Finance)
III. If deviation > 5% → flag as NUMERICAL_ERROR

Layer 4: Self-Correction Loop (Remediation)

I. If Validator finds flaws → return to Analyst with specific correction instructions
II. Maximum 3 correction loops before escalating to user with transparency report
III. Log all correction iterations for benchmarking hallucination rate

Layer 5: Confidence Scoring

I. Each output gets a confidence score (1–10) based on:
   A. Number of corroborating sources (more = higher)
   B. Recency of data (older = lower)
   C. Consensus alignment (contrarian = lower confidence, not lower validity)

Why This Project

1. 2026 Is the Most Complex Financial Era in Modern History

- **AI Spending Bubble Uncertainty**: By early 2026, investors are debating whether AI capex (led by Microsoft, Google, Meta, and Amazon spending $200B+ collectively) will yield returns or become the next dot-com bust. CNBC reports the "AI market is splintering" as investors scrutinize who is spending wisely vs. recklessly. A system that can perform second-order reasoning on AI supply chains is no longer a luxury — it's a necessity.
- **Geopolitical Supply-Chain Disruption**: Ongoing US-China tech sanctions, shifting trade tariffs, and energy policy changes create cascading second-order effects that are invisible to first-order analysis tools. When a tariff changes on rare earth minerals, the impact ripples through semiconductors → data centers → AI companies → energy providers. No existing tool traces this chain.

2. The "Reasoning Gap" Is Widening

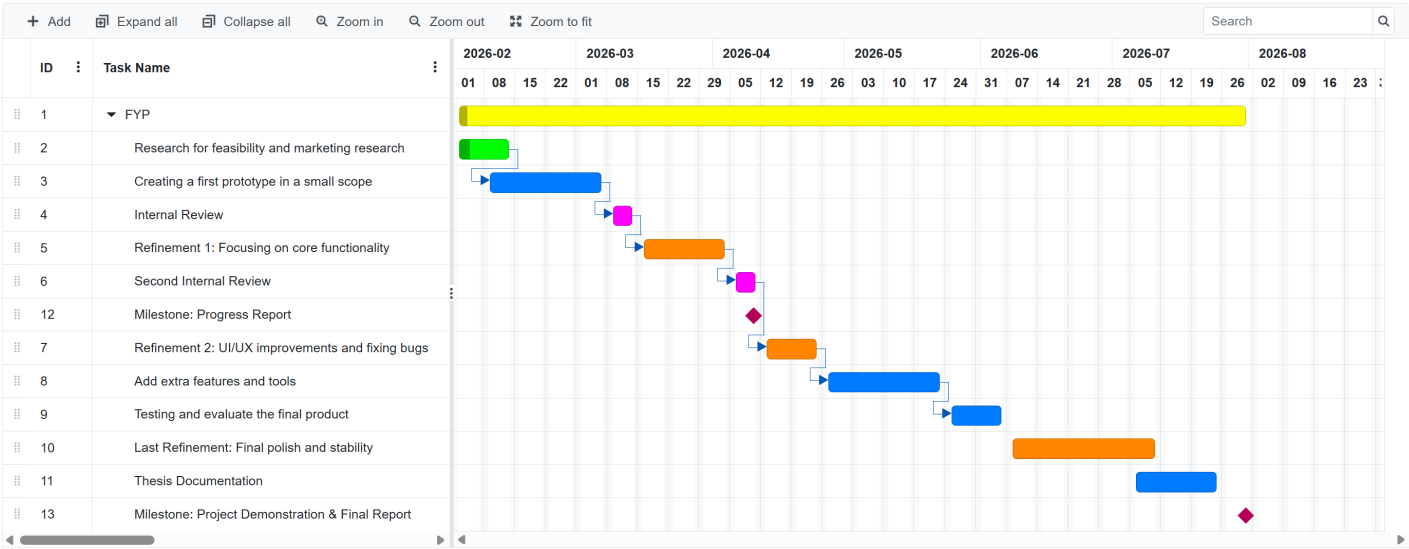
- Most AI tools provide "Nvidia stock rose because they launched a chip", which is a first-order observation anyone can see. What investors need is "Nvidia's new chip will increase power consumption by 40% per data center, which benefits stocks like “Constellation Energy”, whose nuclear plants provide 24/7 baseload power, and their PPA contracts lock in revenue for 10+ years." This is the "Reasoning Gap" my FYP aims to bridge.

3. AI Hallucination in Finance Is a Critical Unsolved Problem

- In 2025–2026, multiple high-profile cases of AI-generated financial misinformation have eroded trust. AI tools that fabricate earnings numbers, cite non-existent SEC filings, or invent analyst opinions pose real financial risk to users who act on them.
- No existing open-source project treats hallucination as a first-class architectural concern. FinGPT and FinRobot generate outputs without validation. My project's 5-layer anti-hallucination pipeline (with RAG grounding, numerical cross-checks, and self-correction loops) positions it as

a pioneering contribution to trustworthy financial AI.

4. Local LLMs Have Finally Reached "Good Enough" Quality

- As recently as 2024, running a capable reasoning model locally was impractical. But by early 2026:
  - DeepSeek-R1 (32B) delivers reasoning quality comparable to GPT-4 for structured analytical tasks
  - RTX 5070 Ti (16GB VRAM) can run Q4-quantized 32B models at practical speeds
  - Ollama has matured into a production-ready local inference server
- This means the "low cost" constraint is newly achievable. A year ago, this project would have required $100+/month in API costs. Today, it can run for $0/month, making it feasible as a student FYP.

Planning Chart